

ReconBoost: Boosting Can Achieve Modality Reconciliation

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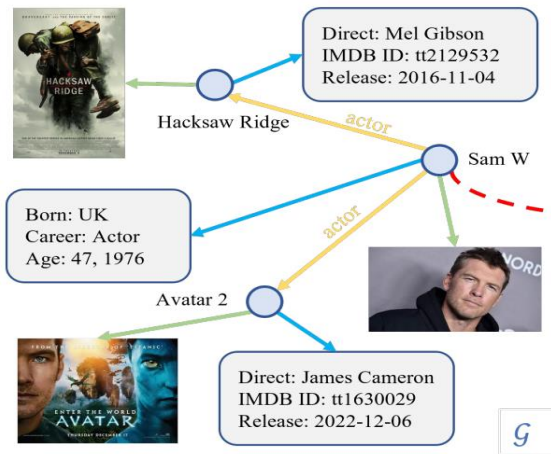
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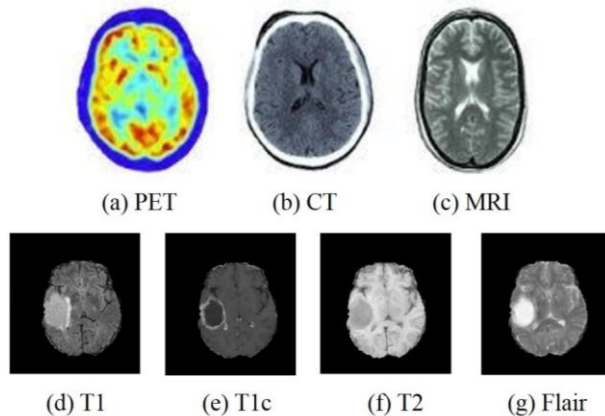
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Background

□ Real-world data usually follows a multi-modal nature



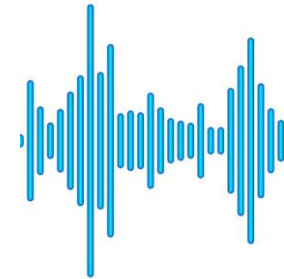
Knowledge Graph



Medical Diagnosis

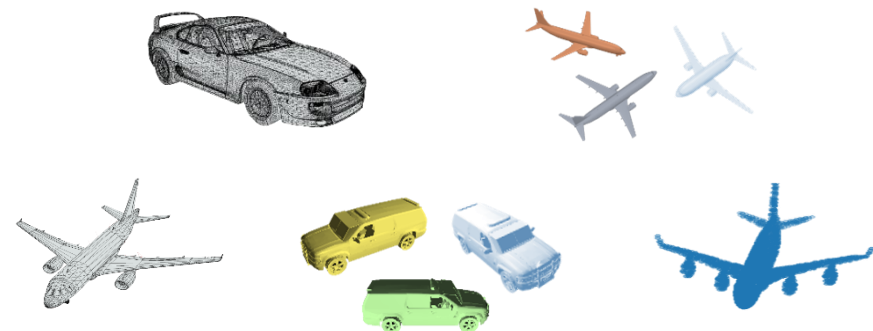


Visual



Audio

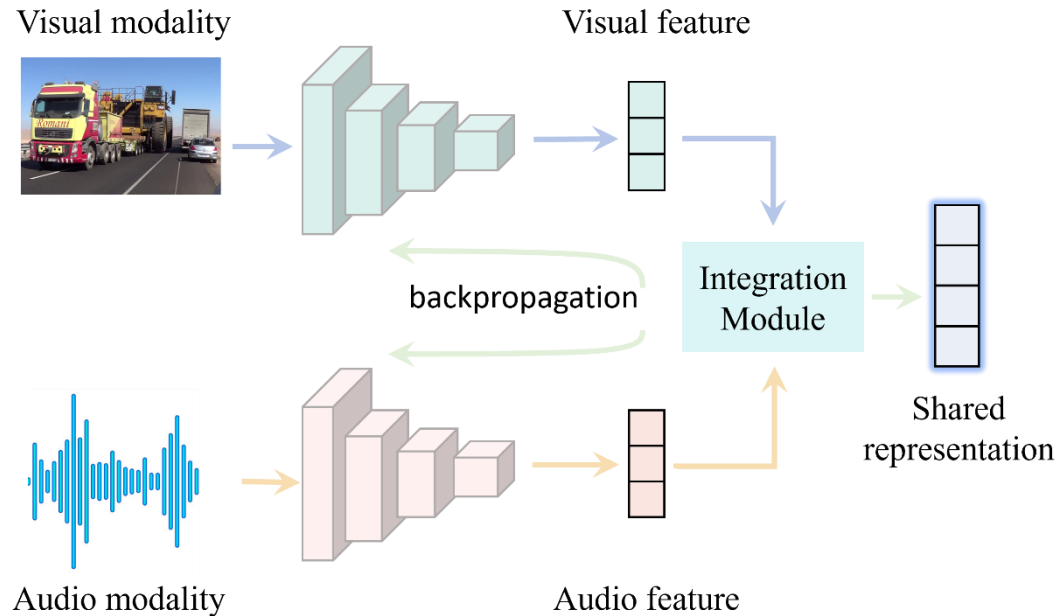
Scene Understanding



Cross-modal Retrieval

Background

□ Multi-modal Joint Learning



- The prevailing paradigm in multi-modal learning typically employs a **joint learning** strategy.
- Various MML studies focus on integrating modality-specific features into **a shared representation** for downstream tasks.

Background

□ Joint Gradient Descent Update

The gradient update rule of k-th modality learner

$$\begin{aligned}\theta_k^{t+1} &= \theta_k^t - \eta \cdot \nabla_{\theta_k^t} \mathcal{L}(\Phi_M^t(x), y) \\ &= \theta_k^t - \eta \cdot \frac{1}{N} \sum_{i=1}^N \left(\frac{\partial W_k \cdot \mathcal{F}_k(\theta_k^t; m_i^k)}{\partial \theta_k^t} \right)^\top \cdot \underbrace{\frac{\partial \ell(\Phi_M^t(x_i), y_i)}{\partial \Phi_M^t(x_i)}}_{\text{shared}}\end{aligned}$$

If $\frac{\partial \ell(\phi_k^t(x_i), y_i)}{\partial \phi_k^t(x_i)}$ **approximates** $\frac{\partial \ell(\Phi_M^t(x_i), y_i)}{\partial \Phi_M^t(x_i)}$, this modality will **converge fast**.

If $\frac{\partial \ell(\phi_k^t(x_i), y_i)}{\partial \phi_k^t(x_i)}$ **does not approximate** $\frac{\partial \ell(\Phi_M^t(x_i), y_i)}{\partial \Phi_M^t(x_i)}$, this modality will **be stuck**.



uni-modal score gradient

Background

□ Modality Competition – Empirical Observation

- The gradient of audio $\frac{\partial \ell(\phi_k^t(x_i), y_i)}{\partial \phi_k^t(x_i)}$ **approximates** $\frac{\partial \ell(\Phi_M^t(x_i), y_i)}{\partial \Phi_M^t(x_i)}$

Audio modality will **converge fast** and **overpower the learning process**.

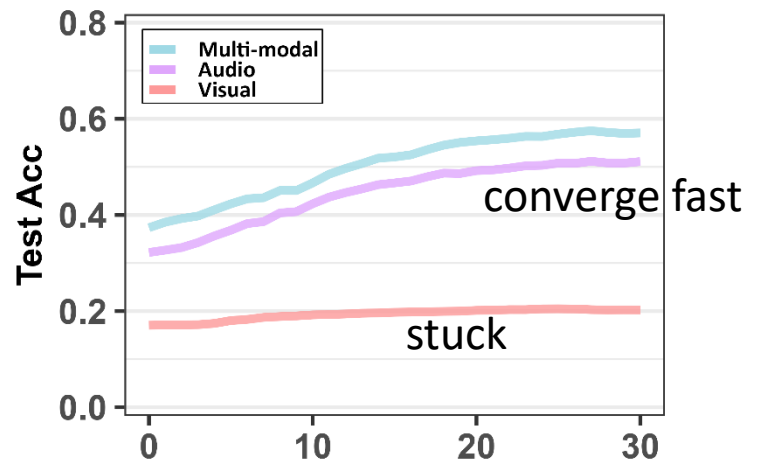
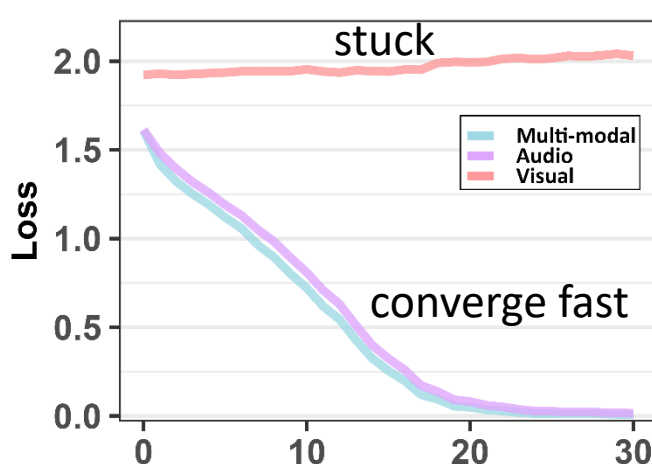
- The gradient of visual $\frac{\partial \ell(\phi_k^t(x_i), y_i)}{\partial \phi_k^t(x_i)}$ **does not approximate** $\frac{\partial \ell(\Phi_M^t(x_i), y_i)}{\partial \Phi_M^t(x_i)}$

Visual modality will **be stuck at bad local optimums**.

dominant
modality



weak
modality



Performance on Audio-Visual dataset

Background

□ Modality Competition – Theoretical Analysis

- The multi-modal joint learning neural network will not efficiently learn all features from different modalities, and **only a subset of modality encoders** will capture sufficient feature representations.

Theorem 4.2. For sufficiently large $K > 0$ and every $\eta \leq \frac{1}{\text{poly}(K)}$, after $T = \frac{\text{poly}(K)}{\eta}$ many iteration, for the multi-modal network $f^{(t)}$, $f^{r(t)} := \mathcal{C}(\varphi_{\mathcal{M}_r}^{(t)})$ w.h.p :

- Training error is zero:

$$\frac{1}{n} \sum_{(\mathbf{X}, y) \in \mathcal{D}} \mathbb{I}\{\exists j \neq y : f_y^{(T)}(\mathbf{X}) \leq f_j^{(T)}(\mathbf{X})\} = 0.$$

- For $r \in [2]$, with probability $p_{3-r} > 0$, the test error of $f^{r(T)}$ is high:

$$\Pr_{(\mathbf{X}^r, y) \sim \mathcal{P}^r} (\exists j \neq y : f_y^{r(T)}(\mathbf{X}^r) \leq f_j^{r(T)}(\mathbf{X}^r)) \geq \frac{1}{K}$$

where $p_1 + p_2 = 1 - o(1)$, and $p_r \geq m^{-O(1)}$, $\forall r \in [2]$.

Modality Competition

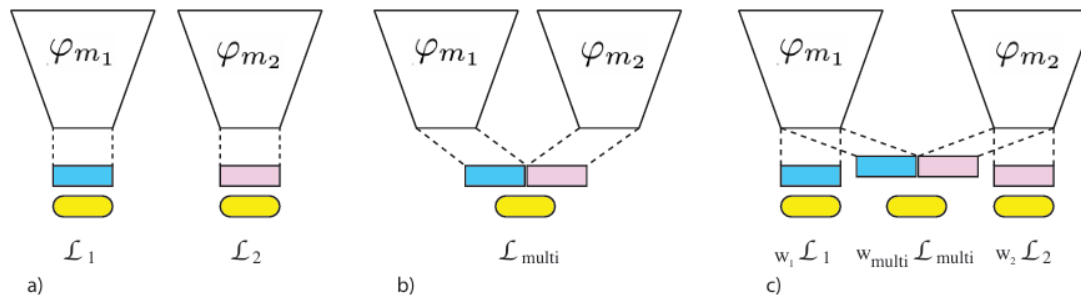
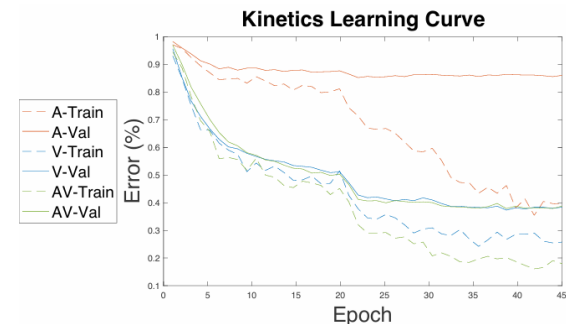
For every $j \in [K]$, at every iteration t , if $\mathcal{M}_{r_j}$ is the winning modality, then $\Gamma_{j, r_j}^{(t)}$ will grow faster than $\Gamma_{j, 3-r_j}^{(t)}$. When $\Gamma_{j, r_j}^{(t)}$ reaches the threshold $\Theta(\beta) = \tilde{\Theta}(1)$, $\Gamma_{j, 3-r_j}^{(t)}$ still sticks at initial level around $\tilde{O}(\sigma_0)$.

Balanced Multimodal Learning

□ Add additional uni-modal loss

- Uni-modal learner perform better in some cases.
- Each modal learner has **its own convergence rate**.
- Add additional uni-modal loss and assign dynamic weights

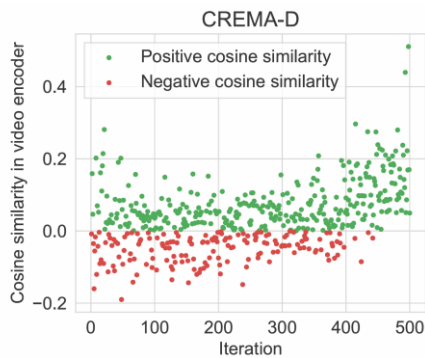
Dataset	Multi-modal	V@1	Best Uni	V@1	Drop
Kinetics	A + RGB	71.4	RGB	72.6	-1.2
	RGB + OF	71.3	RGB	72.6	-1.3
	A + OF	58.3	OF	62.1	-3.8
	A + RGB + OF	70.0	RGB	72.6	-2.6



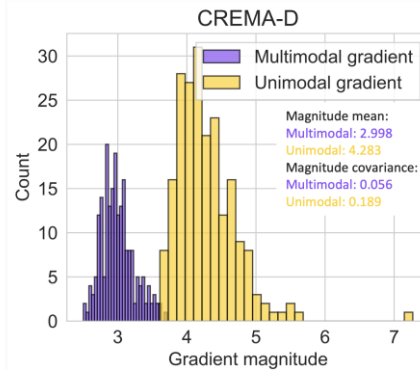
Balanced Multimodal Learning

□ Boosting MML with innocent uni-modal assistance

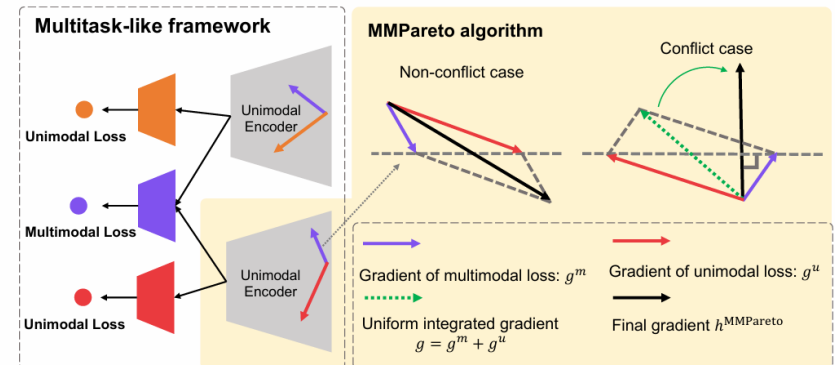
- **Gradient conflict** between multimodal and unimodal learning objectives
- Both gradient magnitude and covariance of MML loss are smaller.
- Analyze **Pareto** integration under the multimodal scenario.



(a) Direction conflict.



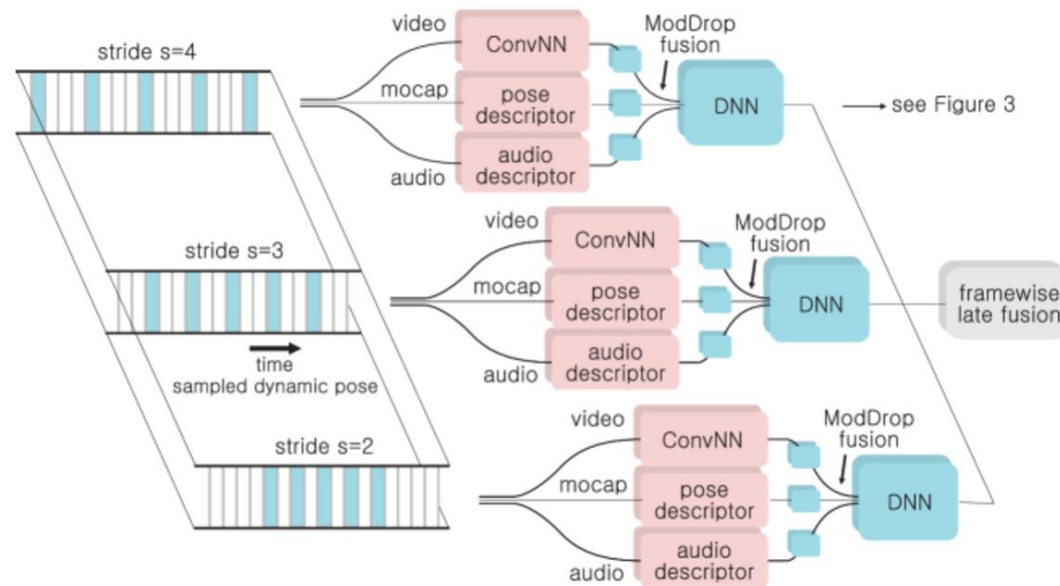
(b) Gradient distribution.



Balanced Multimodal Learning

□ Modality Dropout

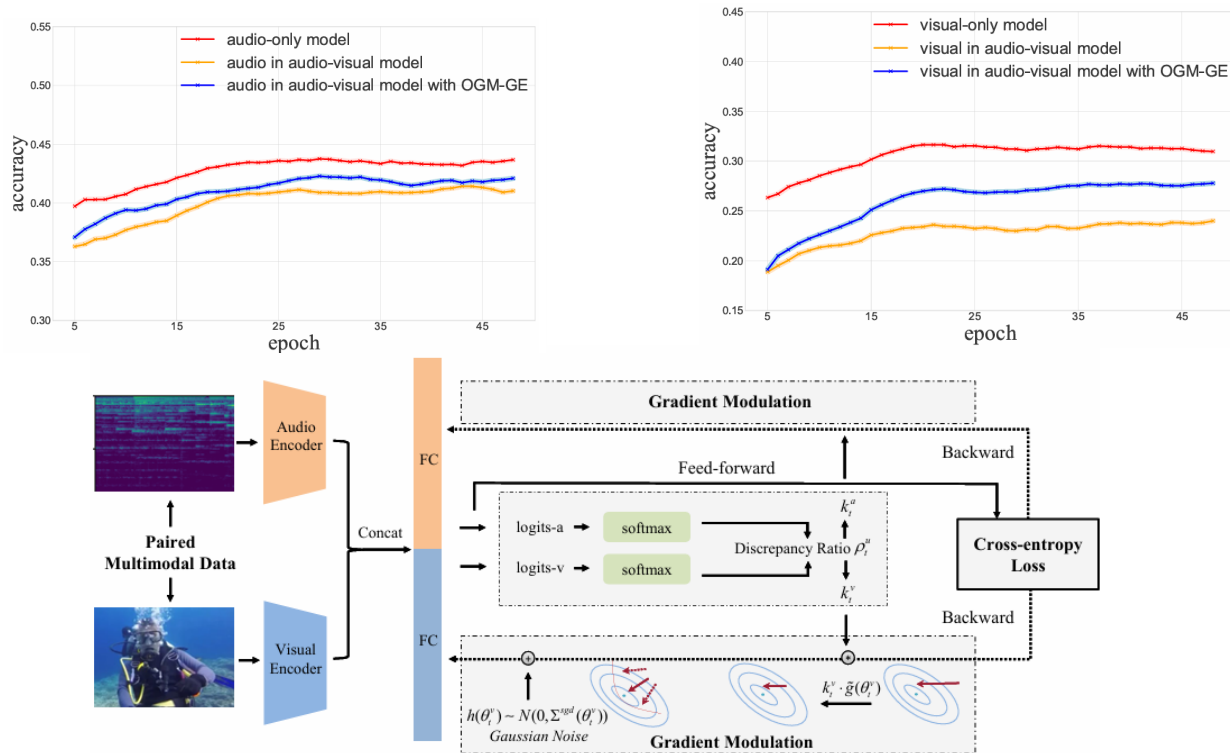
- Drop modalities with the specific probability in the **feed-forward** stage.



Balanced Multimodal Learning

□ On-the-fly Gradient Modulation

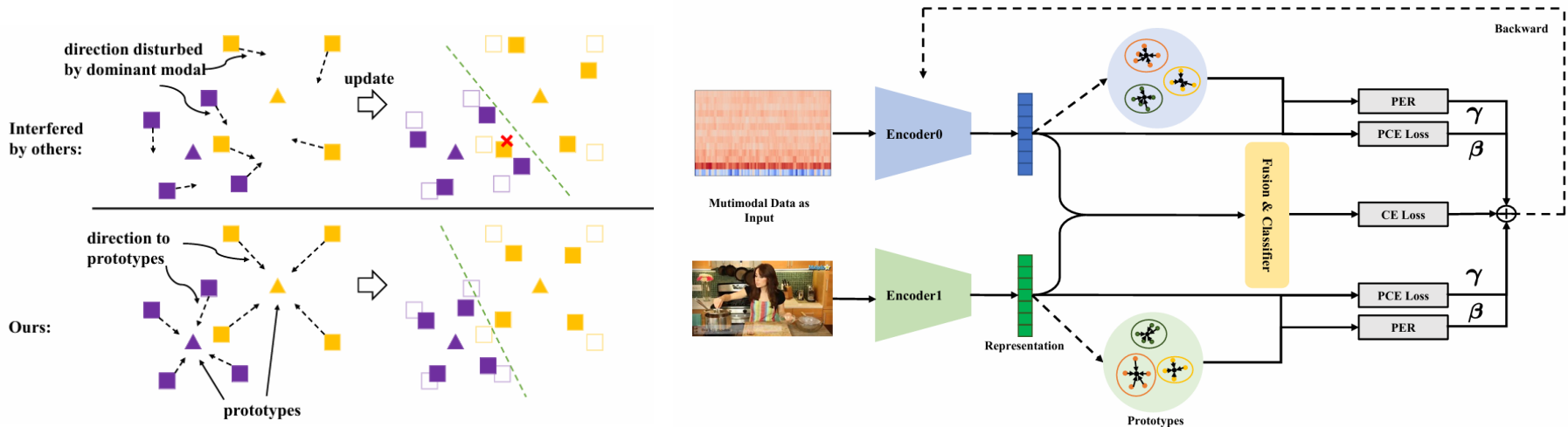
- Control the optimization process of each modality learner **dynamically**.
- Introduce extra Gaussian Noise to enhance generalization.



Balanced Multimodal Learning

□ Prototypical Modal Rebalance for Multimodal Learning

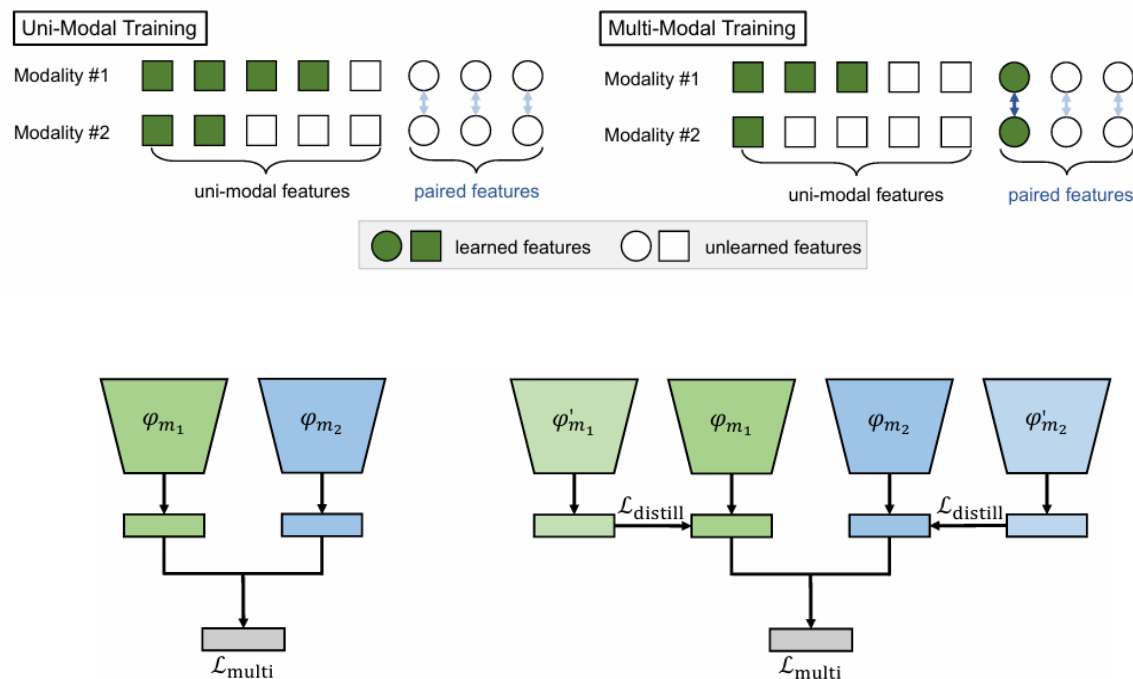
- Slow-learning modality's updating **direction** is severely disturbed by the dominant one
- Propose prototype-based method to adjust updating direction for better uni-modal performance



Balanced Multimodal Learning

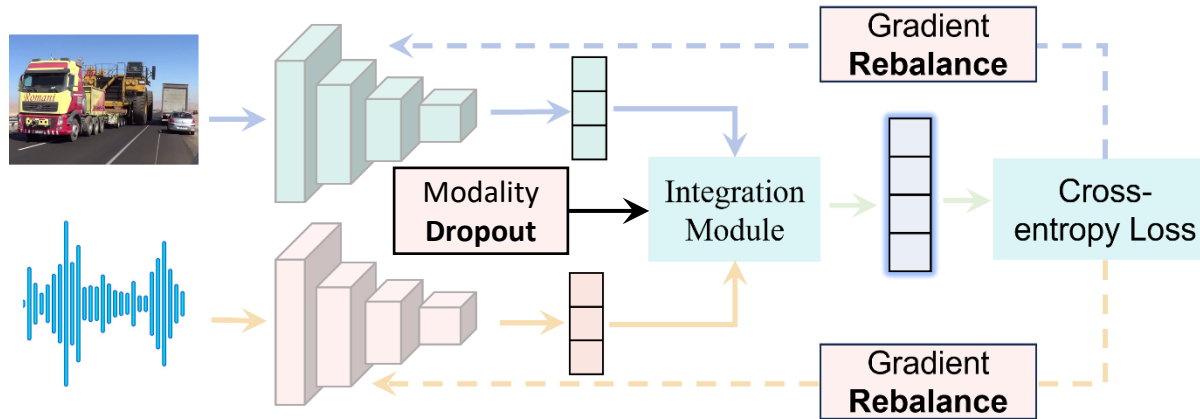
□ Distill knowledge from well-trained uni-modal model

- Recent MML suffer from insufficient learning of uni-modal features on each modality.

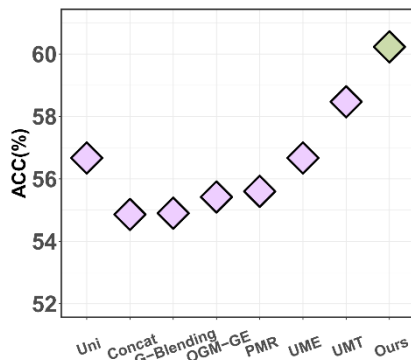


Motivation

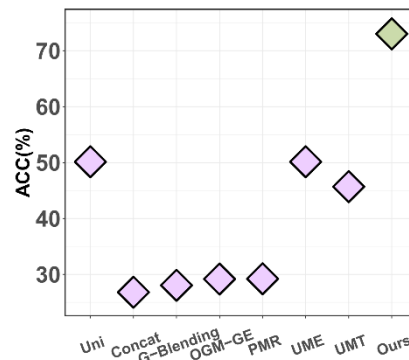
- The primary concern is how to **balance** optimization progress across multi-modal learners.



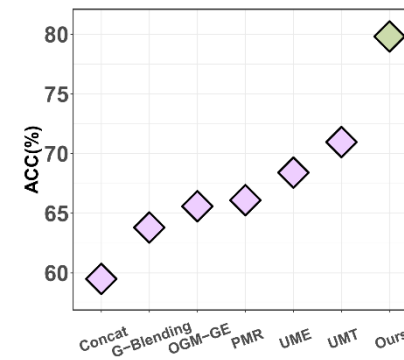
- Given the nature of **joint optimization**, only limited improvements can be achieved.



(a) Audio Modality



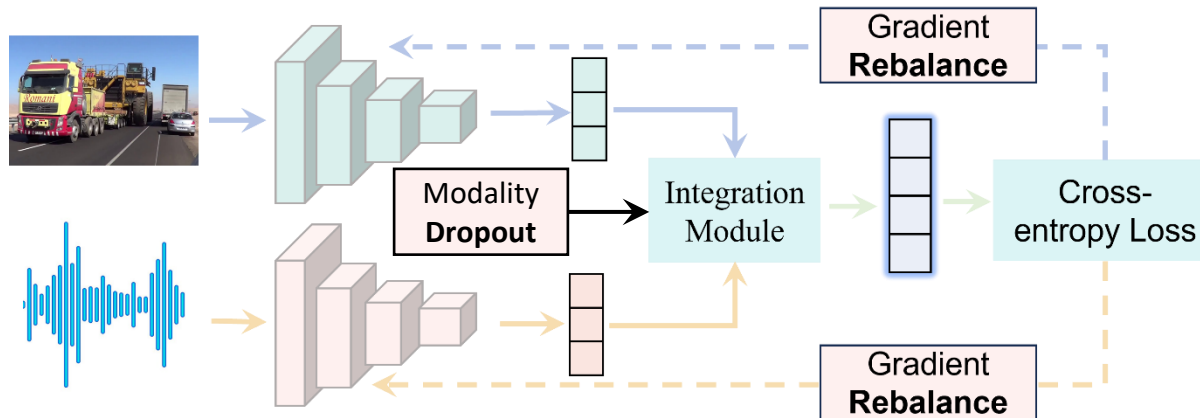
(b) Visual Modality



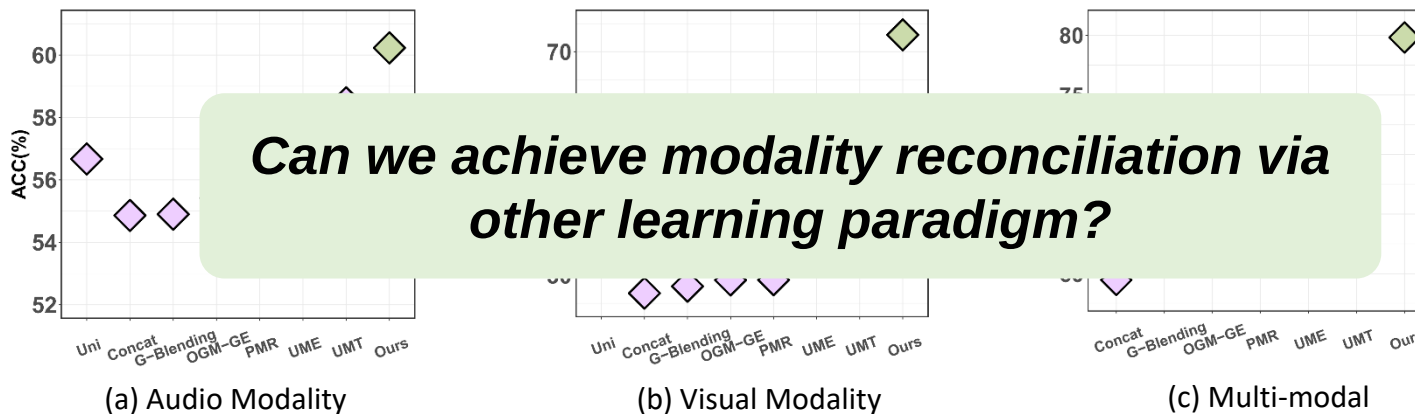
(c) Multi-modal

Motivation

- The primary concern is how to **balance** optimization progress across multi-modal learners.



- Given the nature of **joint optimization**, only limited improvements can be achieved.



Main part

□ Naive version of modality-alternating learning

Step 1: Each time, we pick a specific modality learner ϕ_k to **update**, and keep other fixed.

$$\vartheta_k^{t+1} = \vartheta_k^t - \eta \cdot \nabla_{\vartheta_k^t} \mathcal{L}(\phi_k^t(m^k), y)$$

$$\mathcal{L}(\phi_k^t(m^k), y) = \frac{1}{N} \sum_{i=1}^N \ell(\phi_k(\vartheta_k^t; m_i^k), y_i)$$

Step 2: Multi-modal scores are **merged** to obtain the final score.

$$\Phi_M(x_i) = \sum_{k=1}^M \phi_k(\vartheta_k; m_i^k)$$

- The gradient across different modalities are naturally **disentangled** from each other, alleviating the modality competition issue.
- This approach ensures the exploitation of uni-modal features, but **neglects the investigation of cross-modal interaction**.

Main part

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- The gradient across different modalities are naturally disentangled from each other.

How to design a more effective modality supervised signal?

- The naive version of modality-alternating learning neglects the investigation of cross-modal diversity.

Main part

□ Modality-alternating Update with Dynamic Reconciliation

Step 1: Each time, we pick a specific modality learner ϕ_k to **update**, and keep other fixed.

$$\vartheta_k^{t+1} = \vartheta_k^t - \eta \cdot \nabla_{\vartheta_k^t} \tilde{\mathcal{L}}_s(\phi_k^t(m^k), y)$$

$$\tilde{\mathcal{L}}_s(\phi_k(m^k), y) = \frac{1}{N} \sum_{i=1}^N \left[\underbrace{\ell(\phi_k(\vartheta_k; m_i^k), y_i)}_{\text{agreement term}} - \lambda \cdot \underbrace{\mathbb{D}_s(\Phi_{M/k}(x_i), \phi_k(\vartheta_k; m_i^k))}_{\text{reconciliation regularization term}} \right]$$

Dynamically maintain the trade-off between two items:

- The agreement term aligns the overall predictor with the ground truth.
- The reconciliation regularization term investigates the cross-modal diversity.

Step 2: Multi-modal scores are **merged** to produce the final score.

$$\Phi_M(x_i) = \sum_{k=1}^M \phi_k(\vartheta_k; m_i^k)$$

Main part

□ Connection to the Boosting Strategy

The overall optimization property of ReconBoost is **unclear**

$$\tilde{\mathcal{L}}_s(\phi_k(m^k), y) = \frac{1}{N} \sum_{i=1}^N \left[\underbrace{\ell(\phi_k(\vartheta_k; m_i^k), y_i)}_{\text{agreement term}} - \lambda \cdot \underbrace{\mathbb{D}_s(\Phi_{M/k}(x_i), \phi_k(\vartheta_k; m_i^k))}_{\text{reconcilement regularization term}} \right]$$

Theorem 1. Connection to the Gradient Boosting (GB) method

Let the reconcilement regularization be a KL divergence function:

$$\mathbb{D}_s(\Phi_{M/k}(x_i), \phi_k(\vartheta_k; m_i^k)) = \mathbb{D}_{KL,s}(\Phi_{M/k}(x_i) | \phi_k(\vartheta_k; m_i^k))$$

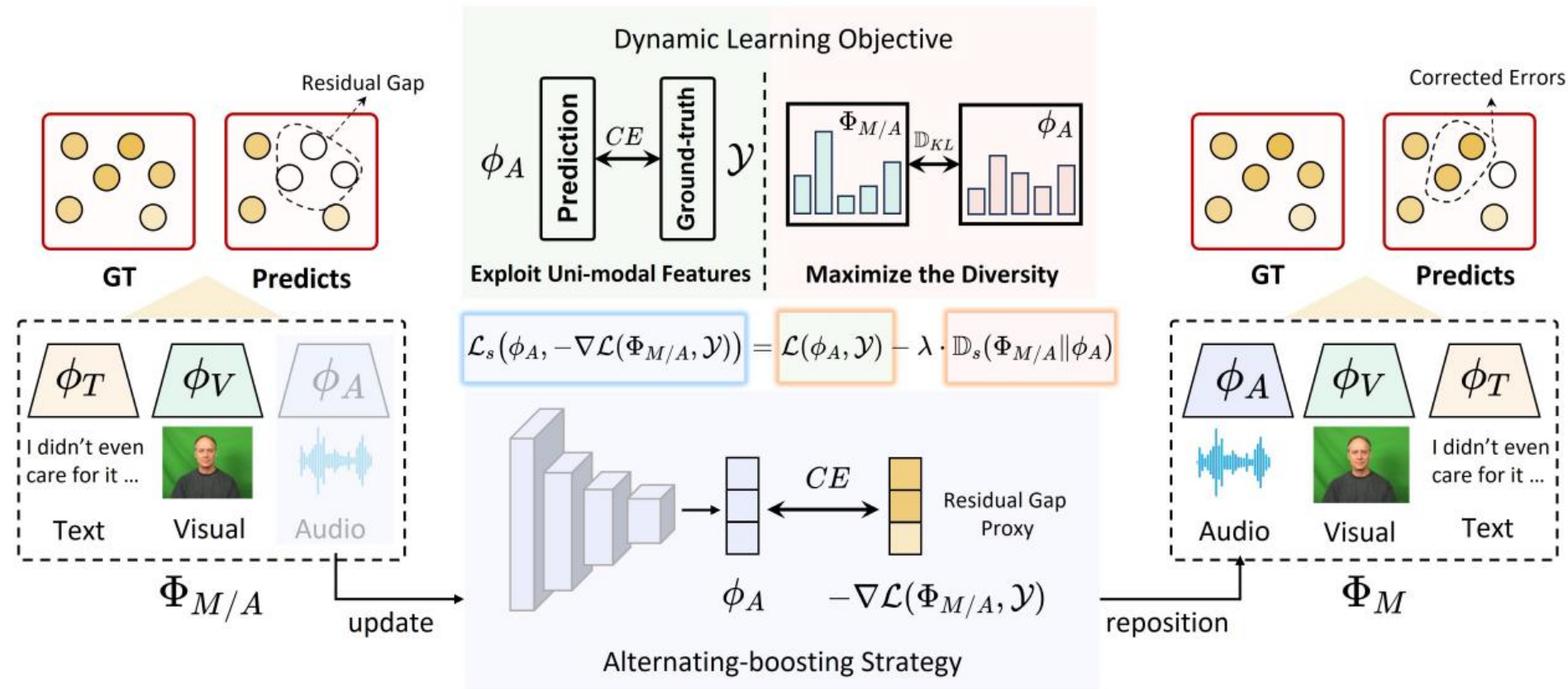
Then,

$$\nabla_{\vartheta_k} \tilde{\mathcal{L}}_s(\phi_k(m^k), y) \iff \nabla_{\vartheta_k} \mathcal{L}(\phi_k(m^k), -\nabla_{\Phi_{M/k}} \ell(\Phi_{M/k}(x), y))$$

- Optimizing the dynamic loss functions $\tilde{\mathcal{L}}$ in ReconBoost **consistently** optimizes the original loss $\mathcal{L}$ with a progressively changing pseudo-label in GB algorithm (Friedman, 2001).
- The updated modality learner can focus on the errors made by others.
- ReonBoost only preserves the last learner of each modality, formulating **alternating-boosting strategy**.

Main part

□ Pipeline of ReconBoost



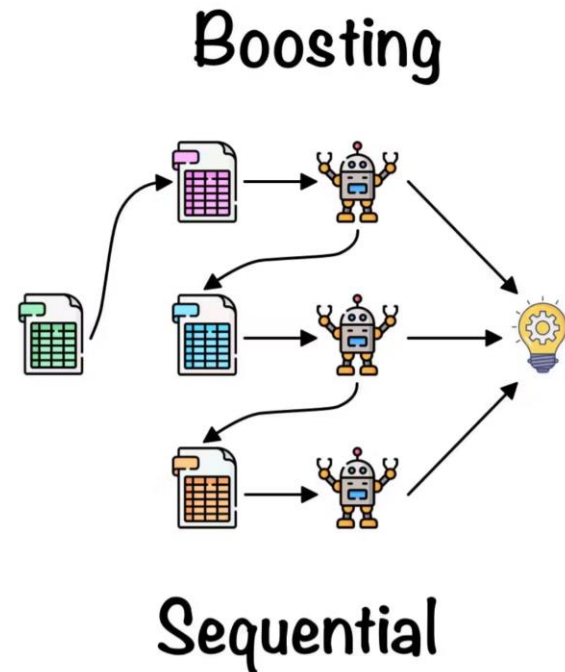
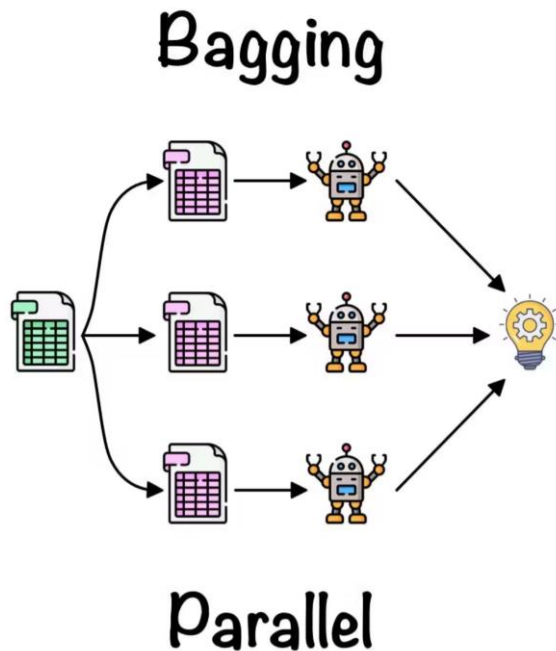
- ReconBoost can realize **an alternating version** of the well-known gradient boosting algorithm.
- ReconBoost purses a **reconciliation** between the exploitation of uni-modal features and the exploration of cross-modal diversity.

In-depth Analysis

❑ Multimodal learning is a form of **Ensemble Learning**.

- Ensemble learning is a machine learning technique that combines multiple models to improve the accuracy of predictions.

❑ Bagging & Boosting



Experiments

□ Quantitative Comparisons

Method	AVE	CREMA-D	MN40	MOSEI	MOSI	CH-SIMS
AudioNet	59.37	56.67		52.29	54.81	58.20
VisualNet	30.46	50.14	80.51	50.35	57.87	63.02
TextNet	—	—	—	66.41	75.94	70.45
Concat Fusion	62.68	59.50	83.18	66.71	76.23	71.55
G-Blending	62.75	63.81	84.56	66.93	76.45	71.55
OGM-GE	62.93	65.59	85.61	66.67	76.01	71.10
PMR	64.20	66.10	86.20	66.41	76.12	70.90
UME	66.92	68.41	85.37	63.88	76.97	71.77
UMT	67.71	70.97	90.07	67.04	75.80	71.55
Ours	71.35	79.82	91.78	68.61	77.96	73.88

Experiments

□ Modality-specific Encoder Evaluation

Method	CREMAD Dataset				AVE Dataset			
	Audio	Visual	MIR	DMC	Audio	Visual	MIR	DMC
Uni-train	56.67	50.14	1.13	-	59.37	30.46	1.95	-
Concat Fusion	54.86	26.81	2.05	1.81	55.47	23.96	2.32	1.19
G-Blending	54.90	28.05	1.96	1.73	55.80	24.12	2.31	1.19
OGM-GE	55.42	29.17	1.90	1.68	56.51	25.52	2.21	1.14
PMR	55.60	29.21	1.90	1.68	57.20	26.30	2.17	1.12
UMT	58.47	45.69	1.28	1.13	60.70	31.07	1.95	1.00
Ours	60.23	73.01	0.82	0.73	61.20	39.06	1.57	0.80

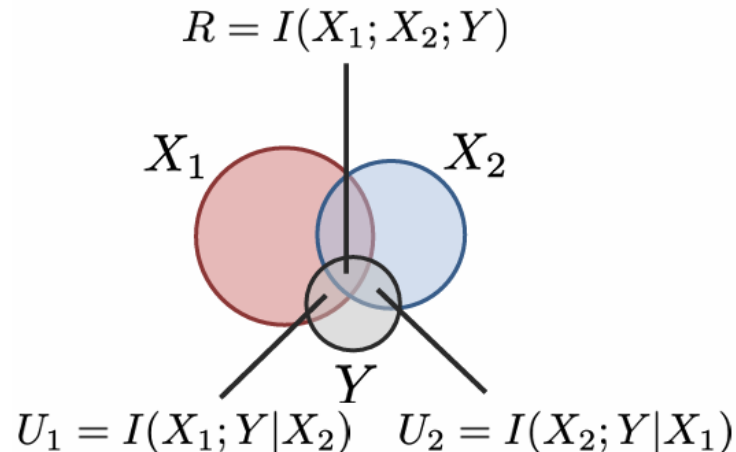
Conclusion

- ❑ **Methodologically:** propose a novel multi-modal **alternating learning paradigm** to address notorious modality competition issue.
- ❑ **Theoretically:** show that by choosing a KL-divergence-based reconciliation term, our proposed method can realize **an alternating version** of the well-known gradient boosting method.
- ❑ **Empirically:** Comprehensive experiments justify the effectiveness of our proposed framework on various multi-modal scenarios.

Limitations & Future work

- ❑ Our work only addresses the modality competition issue in supervised learning
- ❑ Exploring the phenomenon of modality competition in

Multi-modal Contrast Learning



- **Focuses** only on **SHARED** information
- **Ignores** modality-specific **UNIQUE** information

Thanks for your listening!



code